KRISHNA CLASSES

WORKSHEET

SUBJECT- MATHS

CHAPTER- 2

NAME OF STUDENT –

ISSUE DATE –

Chapter-2

1. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is
(a) 10 (b) -10
(c) 5 (d) -5

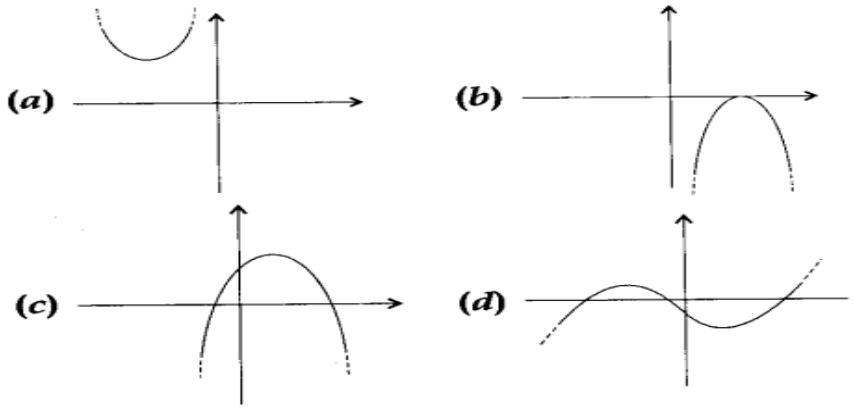
2. If one of the zeroes of the quadratic polynomial $(k - 1)x^2 + kx + 1$ is – 3, then the value of k is

a. $\frac{4}{3}$ c. $\frac{-4}{3}$

b. $\frac{2}{3}$ d. $\frac{-2}{3}$

3. The zeroes of the quadratic polynomial $x^2 + 99x + 127$ are
(a) both positive (b) both negative
(c) one positive and one negative (d) both equal

4. Which of the following is not the graph of quadratic polynomials?



5. A quadratic polynomial, whose zeros are -4 and -5, is
(a) x^2-9x + 20 (b) x^2 + 9x + 20
(c) x^2-9x- 20 (d) x^2 + 9x- 20

6. If the zeroes of the quadratic polynomial $Ax^2 + Bx + C$, C # 0 are equal, then
(a) A and B have the same sign (b) A and C have the same sign
(c) B and C have the same sign (d) A and C have opposite signs

7. If one of the zeroes of the cubic polynomial $x^3 + px^2 + qx + r$ is -1, then the product of the other two zeroes is
(a) p + q + 1 (b) p-q- 1
(c) q – p + 1 (d) q – p – 1

8. The degree of a constant polynomial is:
a) 0 b) 1

c) 2 d) undefined

9. The value of k for which the polynomial x3−3x2+kx−4 has 1 and 2 as its zeroes is:

a) 5 b) 6

c) 7 d) 8

10. A quadratic polynomial, whose zeroes are -3 and 4, is

(a) $x^2 - x + 12$ (b) $x^2 + x + 12$

(c) $\frac{x^2}{2} - \frac{x}{2} - 6$ (d) $2x^2 + 2x - 2$

11. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

(i) x^2-2x-8

(ii) $6x^2-3-7x$

(iii) t^2-15

12. Find a quadratic polynomial each with the given numbers as the sum and product of its zeroes respectively.

(i) 1/4, -1

(ii) 0, √5

(iii) 4, 1

13. If α and β are the zeros of the quadratic polynomial $f(x) = x^2 - x - 4$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$.

14. If α and β are the zeros of the quadratic polynomial f(x)=x2−p(x+1) −c, show that (α +1) (β +1) =1−c.

15. Find the value of 'k' such that the quadratic polynomial $x^2-(k+6)x+2(2k+1)$ has sum of the zeros is half of their product.

16. If the square of the difference of the zeros of the quadratic polynomial $f(x)=x^2+px+45$ is equal to 144, find the value of p.

17. Find the zeros of the polynomial $p(x) = 4\sqrt{3}x^2 + 5x - 2\sqrt{3}$ and verify the relationship between the zeros and its coefficients.

18. If the sum of the zeros of the quadratic polynomial $f(t)=kt^2+2t+3k$ is equal to their product, find the value of k.

19. If one zero of the quadratic polynomial $f(x)=4x^2-8kx-9$ is negative of the other, find the value of 'k'.

20. If α and β are the zeros of the polynomial $f(x)=x2+px+q$, find polynomial whose zeros are $(\alpha+\beta)^2$ and $(\alpha-\beta)^2$.

21. Find the value of 'k' so that the zeros of the quadratic polynomial $3x^2-kx+14$ are in the ratio 7:6.

22. Find the quadratic polynomial where sum and product of the zeros are a and $\frac{1}{a}$

23. Obtain all other zeroes of $3x^4+6x^3-2x^2-10x-5$, if two of its zeroes are $\sqrt{(5/3)}$ and $-\sqrt{(5/3)}$.

24. Find a cubic polynomial with the sum, sum of the product of its zeroes taken two at a time, and the product of its zeroes as 2, –7, –14 respectively.

25. If the zeroes of the polynomial x^3-3x^2+x+1 are a – b, a, a + b, find a and b.

26. If two zeroes of the polynomial $x^4-6x^3-26x^2+138x-35$ are $2 \pm\sqrt{3}$, find other zeroes.

27. Find the value of "p" from the polynomial $x^2 + 3x + p$, if one of the zeroes of the polynomial is 2.

28. Find the value of "x" in the polynomial $2a^2 + 2xa + 5a + 10$ if $(a + x)$ is one of its factors.

29. α and β are zeroes of the quadratic polynomial $x^2 - 6x + y$. Find the value of 'y' if $3\alpha + 2\beta = 20$.

30. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$, $c \neq 0$.